Can one hidden-sector reading cover several dark-matter puzzles? Consistency checks of Ω_c/Ω_b = 3√3 against Planck and DESI DR2, the evolution of the RAR scale, and limits on uniform decay

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Abstract

This record asks whether one reading of dark matter suggested by the author's framework (Mathematical Materiality, MM) can connect several dark-matter puzzles, and confronts the parts that give numbers with public data. The cosmological ratio fit, the evolution of the radial-acceleration-relation (RAR) scale, the decay analysis and the hidden-sector constructions are separate investigations; no joint likelihood or complete dynamical model is presented. Hypothesis (declared, not derived): the effective gravitating cold-dark-matter density represents hidden-sector matter with its own internal interactions — in MM terms the ordinary matter of a «mirror», normal in its own laws, with constants read relative to ours, left generic — with the mean ratio ρ_c/ρ_b = 3√3 imposed separately. It is additionally postulated that hidden radiation does not source our expansion; no covariant implementation is provided.

Can a single reading with a hidden sector cover several dark-matter problems? The 3√3 ratio with Planck and DESI, the galaxy scale over time and the limits on decay

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The archived text is in English.

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Before starting: the words you need

Dark matter. Galaxies and clusters of galaxies move in a strange way. It is as if there were more mass than the mass we see. The history of the whole universe requires it too. This extra mass is called *dark matter*. It does not emit or absorb light: we notice it only through its attraction.

Baryons. Ordinary matter is made of *baryons*. Stars, gas, planets and we ourselves are made of this matter. Baryons are mostly protons and neutrons.

Ω and ω. Ω_b says what part of the universe is made of baryons. Ω_c says what part is made of cold dark matter. The lowercase letters ω_b and ω_c are the same quantities, multiplied by a fixed number. That number is tied to today's expansion. The ratio ω_c/ω_b is equal to Ω_c/Ω_b.

The ratio 3√3. In this work it is imposed that dark matter is, on average in the universe, 3√3 times ordinary matter. 3√3 is about 5.196.

Standard cosmology, ΛCDM. It is the most used model for the history of the universe. It contains dark energy (Λ) and

cold dark matter (CDM, short for *cold dark matter*). *Cold* means that its particles move slowly.

Planck and DESI. Planck is a satellite. It measured the *cosmic microwave background* (CMB): the oldest light of the universe. It set off when the universe was very young. DESI is a large survey of galaxies. It measures *baryon acoustic oscillations* (BAO). They are a typical distance left imprinted in the distribution of galaxies. They act as a cosmic ruler.

Redshift, z. The light of distant galaxies is stretched while the universe expands. The shift is written z. The larger z is, the farther away the galaxy is and the further back in time we see it.

The RAR and a_0. In disc galaxies two accelerations are compared at each point. The *observed* one, g_obs, comes from the motion of stars and gas. The one of the baryons alone, g_bar, comes from Newton's law applied to the matter we see. The link between the two is called the *radial acceleration relation*, RAR. It has a single scale, a_0. For nearby galaxies it is about 1.20×10^{-10} m s^{-2}: it is the SPARC value used in this work.

Fitting and the likelihood. To *fit* a model to the data means to choose its numbers. They are chosen so that the model follows the data as well as possible. The *likelihood* measures how well it succeeds. Here two error scores are used: χ^2 (read *chi-squared*) and Q. The lower they are, the better the model follows the data.

σ. It is the typical error of a measurement.

Registering. A hypothesis is *registered* when it is written down and dated before seeing the data with which it will be tested.

In short

Mathematical Materiality (MM) is the author's framework. This record asks whether a single reading of dark matter, suggested by MM, can address several problems together. The parts that give numbers are compared with public data.

The parts are separate investigations. There are four: the fit of the cosmological ratio, the fit of the evolution of the RAR, the decay analysis and the hidden-sector constructions. **There is no joint likelihood, nor a complete dynamical model that connects them all.**

The hypothesis, stated (section 6).

- The density of cold dark matter that we feel through gravity would be matter of a *hidden sector*. This matter has its own internal interactions. It is different from the visible baryons counted in ω_b.
- In MM terms it is the ordinary matter of a "mirror", normal in its own laws, whose constants are read relative to ours.
- The masses of its particles, its dimensionless couplings and its thermal history remain unspecified. The *couplings* say how strong the forces are; *dimensionless* means that they are pure numbers, without units.
- Separately, the ratio of the cosmological mean densities is imposed: ρ_c/ρ_b = 3√3.

An example from everyday life, to get an idea. Two radios tuned to different frequencies do not disturb each other: one does not hear the other. But both weigh on the same table. Here the hidden sector and ours do not *feel* each other through light, but they attract each other through gravity.

Three clarifications, which the text gives at once:

- **One more postulate.** The interpretation also postulates that the hidden *radiation* does not act as a source of our cosmological expansion. This does not follow from the fact that the two lights are separate. There is no complete, *covariant* model that implements it. Covariant means written in a way that is valid for every observer, as relativity requires.
- **An identical copy is excluded.** Suppose that gravity acts in the same way on everything. Then a copy of our sector at the same temperature would add too much radiation: ΔN_eff ≈ 7.4. This is excluded by the measurements. Collisions, cooling and discs of the hidden sector must be studied separately.
- **One example is not enough.** A worked example, with a single scale factor, satisfies two approximate *screening* inequalities (scattering and residual ionisation). It does not show that such a hidden sector can really work. A complete calculation is missing.

Consistency checks against data. The program `python riproduci.py` reports 745 successful checks. They are algebraic identities, numerical calculations and consistency checks on archived cosmological results. The minimisations

of the cosmological likelihood are not repeated by the program.

1. **The ratio Ω_c/Ω_b = 3√3 = 5.196.**
2. Flat ΛCDM is used, with the sum of the neutrino masses Σm_ν = 0.06 eV.
3. ω_cdm = 3√3·ω_b is imposed. The data are those of Planck 2018 (lite TTTEEE, low-ℓ TT and EE) plus DESI DR2 BAO. The cost in the score Q is T_B = 0.17. It costs 0.23 if the Planck lensing reconstruction is added. The calculations are done with the programs CLASS and Cobaya.
4. Such a small cost means that the imposed ratio agrees with the data.
5. If the ratio is left free, the fits give ω_c/ω_b = 5.22 and 5.21.
6. It is a *retrospective* consistency assessment: it uses data that already existed. It is not an independent *prospective* test, that is, a test on data still to come. A prospective test on specific future data is registered.
7. **The acceleration scale of the RAR grows with redshift in the four bins of MUSE-DARK III.** A *bin* is a group of galaxies at similar distances.
8. Among three laws (a constant, the Hubble rate, the inverse of the cosmic age) H(z) gives the lowest χ^2.
9. How good this fit is, in absolute terms, depends strongly on how the error bars of the plot are read. If the bars are 1σ, $\chi^2 = 5.48$. If they are 95 %, $\chi^2 = 21.04$. In both cases there are three degrees of freedom.
10. Other simple forms do just as well. The galaxies at $z \approx 2$ are in tension with this law.
11. H(z) is kept as a *phenomenological* hypothesis (chosen because it describes the data) and is registered for future data.
12. **No preference for a decay** at a constant rate of (almost) all dark matter into dark radiation. In the computed profiles the lowest value is at Γ = 0, that is, without decay. The conventional threshold is ΔQ = 2.71. There the nominal and provisional limits on the *comoving* loss are about 2.6 % with the lensing reconstruction. Without it, they are 3.8 %. *Comoving* means counted in a volume that grows together with the universe: so the simple dilution due to the expansion does not count as a loss.
13. **Clusters of galaxies.** The fixed ratio gives a baryon fraction, computed on baryons plus cold dark matter, of 0.161. It agrees, roughly, with the baryon budgets of massive clusters. The RAR of galaxies, however, does not describe clusters (X-COP): that problem remains.

Problems this reading tries to connect (section 7). There are four. The first is the ratio between dark matter and baryons. The second is the cosmological scale of a_0. The third is the absence of direct-detection signals. The fourth is the behaviour of dark matter: in cosmology it behaves as *cold*, in galaxies it follows the RAR. The connections have different statuses: imposed numerical hypotheses, phenomenological fits, qualitative assumptions. Other research programmes also address combinations of these problems. The case for the reading rests on this *consilience*: a single idea that holds many together. And it grows only as more connections become quantitative.

How this work can turn out wrong (section 10). There are two quantitative hypotheses, with failure thresholds declared in advance for future tests.

What remains open.

- The dynamics of the hidden sector: thermal history, plasma, collisions.
- The Bullet cluster in detail. It is a pair of clusters that have collided.
- a_0 obtained from the theory. The literal value of MM, $cH_0/2$, is 2.73 times the measured one.
- A derivation of the RAR.
- A covariant model in which hidden radiation does not gravitate.

1. What a physicist will recognise

These parts use tools and ideas that are already known.

- **ΛCDM fits with a fixed ratio ω_c/ω_b.** Standard likelihoods and programs are used: Planck 2018, DESI DR2, CLASS, Cobaya. Fixing the ratio is a single linear constraint between two parameters.
- **The radial acceleration relation** (McGaugh, Lelli and Schombert 2016): $g_{obs} = g_{bar}/(1 - \exp(-\sqrt{(g_{bar}/a_0)}))$. And the question of whether it changes over time (MUSE-DARK III; Alcock 2026). That a_0 is close to $cH_0/2\pi$ is an old remark by Milgrom.
- **Dark matter that decays into dark radiation.** In the CLASS program it is called «dcdm». Audren and colleagues 2014, Poulin and colleagues 2016, Alvi and colleagues 2022, Bucko, Giri and Schneider 2023 have written about it.
- **Hidden or mirror sectors whose ordinary matter appears as dark matter:**
- Kolb, Seckel and Turner 1985;
- mirror matter (Berezhiani, Foot, Ciarcelluti);
- the manyfold universe (Arkani-Hamed, Dimopoulos, Dvali and Kaloper);
- mass rescaling in Randall-Sundrum geometries;

- atomic dark matter (Kaplan and colleagues 2010; Cyr-Racine and Sigurdson 2013).

2. The ratio 3√3

The setup.

- **The model.** Flat ΛCDM. One neutrino with a mass of 0.06 eV and N_ur = 2.0328. This gives N_eff ≈ 3.046, the effective number of neutrino species. It is kept like this for comparison with the CAMB program. Halofit is used, a recipe for the growth of structures on small scales.
- **The parameters.** Both fits vary ln(10^{10}A_s), n_s, 100θ_s, ω_b, τ and A_planck. The *free* model also varies ω_cdm. The D-B model instead imposes ω_cdm = 3√3·ω_b.
- **The data.** Two combinations:
- PD = Planck 2018 lite TTTEEE + low-ℓ TT + low-ℓ EE + DESI DR2 BAO (13 points);
- PDL = PD + the Planck 2018 lensing reconstruction. *Lensing* is the bending of the background light by the matter it meets.
- **The statistic.** Q = Σχ² + ((A_planck − 1)/0.0025)². The statistic to be used was written down before the runs, with a dated hash. It is T_B = Q(D-B) − Q(free): how much the score gets worse if the ratio is imposed.
- **The bands, fixed beforehand.** T_B ≤ 4: compatible. 4 < T_B < 9: tension. T_B ≥ 9: negative outcome in the declared model.
- **The free reference.** It is the refined profile point at Γ = 0 of section 4. It is a run with decaying dark matter, but with the decay switched off. It is equivalent to ΛCDM within 0.011 in Q.

data	Q found, D-B	Q found, free	T_B	free ω_c/ω_b
PD	1018.007	1017.837	**0.17**	5.221
PDL	1027.828	1027.600	**0.23**	5.207

How to read it. With the PD data, imposing the ratio raises Q from 1017.837 to 1018.007: T_B = 0.17. It is far below 4, so in the *compatible* band. Left free, the ratio comes out at 5.221, close to 5.196.

The cautions.

- **The values are numerical estimates from the minima found, not certified global minima.** A program that looks for the minimum can stop in a minimum that is not the lowest of all.
- The two starts of the PD constrained fit differ by 0.30 in Q, more than the reported penalty.
- The lowest free ΛCDM minima found in other runs (`corse_6/`: L0_PDL; DD_PD, which ended at Γ = 0) give T_B = 0.26 and 0.24.
- **Where the cost comes from.** The high-ℓ part of Planck (the fine details of the background radiation) gets worse by 0.9-1.5. DESI instead improves by 0.5-0.7.
- **A cross-check.** An earlier fit with CAMB (Planck + DESI DR2) gave +0.04.
- **The limits.**
- Σm_ν was fixed.
- The uncertainty of the free ratio is not quantified here.
- Nothing here derives 3√3.
- **The registration for future data** is in the package (section *How the work was done*).

3. The RAR scale and redshift

The data.

- The four bins of MUSE-DARK III, put into a table by Alcock (2026). They were *digitised* from figure 3 of Ciocan and colleagues (2026). This means that the values were read from the plot, point by point.
- Today's anchor point, from SPARC (a collection of nearby galaxies): 1.20 ± 0.02 ± 0.24. The amplitudes are in units of 10^{-10} m s^{-2}.
- **The problem of the bars.** The plot does not say explicitly what confidence level the grey error bars have. The source uses 95 % intervals for the fit on the full sample, and applies the same procedure to the bins. So both readings are kept: c68 (bars as 1σ) and c95 (bars as 95 %). Neither of the two is chosen on the basis of how well it fits.

The treatment. A flat universe is used, with Ω_m = 0.315 and Ω_Λ = 0.685. The χ² treat the digitised bins as

independent. The bins also have uncertainties in common, of modelling and of calibration. This approximation does not fully represent them.

model and fitted parameters	c68: parameters	c68: χ^2 (degrees of freedom)	c95: χ^2
constant: A	A = 2.279	28.83 (3)	110.8
Hubble rate, $A \cdot H(z)/H_0$	A = 1.390 ± 0.032	5.48 (3), p = 0.14	21.04
inverse age, $A \cdot t_0/t(z)$	A = 1.069	15.33 (3)	58.9
power, A(1+z)^γ	γ = 0.78 ± 0.15 (0.83 ± 0.13 with SPARC)	1.16 (2)	4.5

How to read it. With the c68 reading the constant has $\chi^2 = 28.83$: it does badly. The Hubble rate has $\chi^2 = 5.48$: it does well. The inverse age has 15.33. The power law has 1.16, but it has two parameters instead of one.

What it shows.

- The four digitised bins favour a scale that grows.
- Among the three laws with a single parameter, H(z) gives the lowest χ^2. The ΔAICc criterion is 9.85 in favour of H over the inverse-age law. The models, however, are not *nested*: one is not a special case of the other. So this number does not convert into σ.
- With the c95 reading, H(z) fits badly, even if better than the other two. The *relative* preference (which one does better) and the *absolute* adequacy (whether it really does well) are kept separate.

What it does not show.

- **H is not the only one.** Forms chosen after seeing the data do just as well or better: (1+z)^¾ with χ^2 1.20, or 1 + ln(1+z) with 2.39.
- **H is not a power law.** Its effective logarithmic slope is 0.84 between z = 0 and 1, and 1.11 between the extreme bins.
- **The rescaling.** Multiplying all values of a_0 by the same number, equal at every z, changes the amplitude but not the evolution. A common systematic error on stellar masses, however, does not necessarily propagate exactly like this.
- **H and the age have close shapes.** Once the amplitudes are fitted, the two laws differ by only 0.062 dex between z = 0.5 and 2.5. (A dex is a factor of ten on a logarithmic scale.)

Where there is tension.

- At z ≈ 2 the H law with A = 1.39 gives $a_0 = 4.21$. Milgrom (2017) discussed the galaxies of Genzel and colleagues (2017). With this a_0 those galaxies would have dark-matter fractions well above the measured ones. It is an illustrative check.
- We look at the baryonic Tully-Fisher relation in KMOS3D. At equal circular velocity, the baryonic mass is higher at z ≈ 2.3 than at z ≈ 0.9. With the law $V_f^4 = G M_b a_0$, an a_0 that grows would do the opposite. It is a qualitative tension, not a direct measurement of how a_0 changes.

The link with MM 2.42. MM 2.42 had registered the inverse-age law. With the stated assumptions on the errors, this law fits these bins worse than H(z). The original registration remains unchanged. Its reassessment is in version 2.42.1 of that record. Here H(z) is kept as a phenomenological hypothesis, to be tested in the future.

4. Limits on uniform decay

The model.

- Almost all cold dark matter *decays*, that is, it turns, at a constant rate Γ, into dark radiation. In the CLASS program it is the dcdm model. A very small part, $\omega_cdm = 10^{-5}$, stays stable for technical reasons. It is needed for the *synchronous gauge*, a way of writing the equations.
- The comoving fraction lost between t_i and today is $F = 1 - \exp[-\Gamma(t_0 - t_i)]$. The table uses $t_i \approx 0$ and t_0 from CLASS. Taking t_i at recombination changes none of the rounded limits.

- The *lifetime* is Γ^{-1}.
- **The decay products gravitate.** So this analysis does not limit a disappearance into a component that, by hypothesis, does not contribute to the shared gravitational field.

The profile. Γ is fixed and the best values of all the other parameters are searched for. There are two starts near the best point. The search method is called BOBYQA, with rhoend 0.001.

Γ (km s^{-1} Mpc^{-1})	loss	ΔQ, PDL	ΔQ, PD
0	0	0	0
0.5	0.7 %	0.36	—
1.0	1.4 %	1.49	1.00
1.5	2.1 %	2.21	—
2.01	2.8 %	2.92	2.12
3.0	4.1 %	4.56	2.88

How to read it. The more dark matter decays (larger Γ, larger loss), the worse the score gets (ΔQ rises).

The result.

- **No preference for a decay at a constant rate.** In the profiles the lowest value is at $\Gamma = 0$.
- **The limits.** The conventional threshold $\Delta Q = 2.71$ is used, with a monotone cubic interpolation between the points. The nominal limits are:
- PDL: $\Gamma < 1.86$, loss < 2.6 %, lifetime > 525 Gyr (Gyr means billions of years);
- PD: $\Gamma < 2.74$, loss < 3.8 %, lifetime > 357 Gyr.
- **These limits are provisional.** Their *frequentist* coverage has not been calibrated: it has not been checked how many times, repeating the measurement, the interval would contain the true value. And the numerical convergence of the profiles is incomplete. In 6 points out of 10 the two starts differ by more than the planned tolerance of 0.1 in Q. The difference goes up to 0.54. A shift of 0.3 in ΔQ moves the limit by about 0.3 points of loss (PDL) and by 0.6 (PD). It is a *sensitivity*, not an error bar.

The validations.

- ΛCDM and the decaying model, with $\Gamma = 0$, agree within 0.011 in Q.
- The comparison between the linear calculation and halofit changes $\Delta Q(2.01) - \Delta Q(0)$ by -0.04 at the tested points. It is a sensitivity check. It does not prove that the nonlinear recipe works for the decaying model.
- CLASS and CAMB, with ΛCDM, differ by about 0.35 in Q. The shifts in ω_b, ω_c and H_0 are below 0.27 times the reference uncertainties.

The context. The nominal limits on the lifetime are numerically larger than the published ones. Alvi and colleagues (2022) find 246 Gyr. Bucko, Giri and Schneider (2023) find 288 Gyr. But the data, the likelihoods and the statistical methods are different. For this reason one cannot say which constraint is stronger.

5. Clusters of galaxies

- **The constraint concerns the mean mass densities.** The imposed relation, $\rho_c/\rho_b = 3\sqrt{3}$, concerns the mean densities of the whole universe. On its own it does not fix a ratio between numbers of particles. It does not fix a ratio between densities at a precise place. And it does not give a separate mass budget for each *halo*. The halo is the cloud of dark matter around a galaxy or a cluster.
- **The baryon fraction.** Massive neutrinos are not counted. Then the fraction of baryons, over the total of baryons and cold dark matter, is $1/(1 + 3\sqrt{3}) = 0.161$.
- **X-COP.** It is a group of well-studied clusters. We look at the subsample of five systems with stellar masses (Eckert and colleagues 2022). For four systems the *hydrostatic* baryon fractions within the radius R_{500} go from 0.141 to 0.165. Hydrostatic means computed assuming that the hot gas is in equilibrium. The cluster A2319 gives 0.185. It is flagged as strongly affected by *hydrostatic bias*. It is a rough comparison. No likelihood analysis separates the fixed ratio from the ΛCDM value, 0.157.

- **X-COP also finds that the RAR of galaxies does not describe the acceleration relation in clusters.** That the total baryon budget works out does not solve this problem, which is a separate one.
- **The halos of galaxies alone are not enough.** In clusters the gas fraction is f_gas = 0.12-0.14 and that of stars is f_* = 0.015-0.025. Then for each unit of stellar mass 33-58 units of dark mass are needed. The estimate from the RAR, for a spherical galaxy within 100 kpc, gives only 4.9-16. It holds for stellar masses between 3×10^{11} and 3×10^{10} M⊙. M⊙ is the mass of the Sun. So summing the contributions of the inner parts of galaxies is not enough. This, however, does not determine how the whole mass of the cluster is distributed.

6. The hypothesis and the MM reading (interpretation)

The stated hypothesis.

- We feel cold dark matter through gravity. Its density would represent matter of a hidden sector, with its own internal interactions. This matter is different from the visible baryons counted in ω_b.
- The masses of its particles, its dimensionless couplings and its thermal history remain unspecified.
- Separately, ρ_c/ρ_b = 3√3 is imposed.
- In MM terms the hidden sector is a "mirror", normal in its own laws, whose constants are read relative to ours. Which constants are different, and how, is left generic in this record.

Why it cannot be an identical copy.

- Suppose that gravity acts in the same way on everything. Then a copy of the Standard Model at the same temperature as our sector is incompatible with the limits on the radiation density.
- In the young universe electrons and positrons annihilated, that is, they turned into photons: light of very high energy, gamma rays. After this electron-positron annihilation, the photons and neutrinos of the copy would contribute ΔN_eff ≈ 7.4. ΔN_eff measures the extra radiation compared with the expected one.
- Identical constants, however, do not force identical temperatures.
- And this argument does not apply directly to the selective coupling of radiation postulated below.
- Collisions, cooling and a possible formation of discs require separate modelling.

The example of section 9. It satisfies two approximate screening inequalities, for scattering and for residual ionisation. It does not show a hidden sector that works in cosmology and in astrophysics. The ΛCDM checks with the fixed ratio test that parametrisation, not all the properties of the generic hidden sector.

What is shared and what is not.

- The hidden light and the internal forces of the hidden sector do not act on us.
- The reading also postulates that hidden radiation does not act as a source of our cosmological expansion. Electromagnetic separation alone does not imply this property. Here there is no covariant gravitational model that implements this selective coupling.

How much and where.

- The amount is constrained by the mean ratio 3√3. The bond does not say how it is distributed in space.
- In the cosmological fits the gravitating component is represented by the normal stable cold dark matter. This is a *restricted* ΛCDM fit. It does not show that the proposed hidden sector reproduces the required dynamics. It does not show it for the mean universe, nor for its *perturbations*. Perturbations are the small differences in density from which structures are born.

The RAR as two readings of a single local object.

- Take a baryonic mass with spherical symmetry. Three quantities are defined: L = 2GM_b(<r)/(c²r), S = r/R_H and a_MM = c²/(2R_H). Here M_b(<r) is the baryonic mass inside the radius r. R_H is the *Hubble radius*: the distance at which, by Hubble's law, the expansion of space moves galaxies away at the speed of light.
- With these definitions L/S = g_bar/a_MM.
- The argument of the empirical RAR is $y = g_{bar}/a_0 = (a_{MM}/a_0)(L/S)$. Today (z = 0), with the adopted H_0 and the SPARC normalisation, $a_{MM}/a_0 = 2.73$.
- The extra fraction of the total radial field is exp(−√y), that is, the extra part divided by the total field.
- MM interprets the contribution of the baryons and the extra one as two readings of the same object. This interpretation does not supply a dynamical mechanism for the RAR. And it does not remove the gravitational interaction assumed in the cosmological calculation. A quantitative description of the shape of the RAR and of its scatter remains open.

The form of the density.

- Take a concentrated spherical baryonic mass and look outside it, in the deep regime of the RAR. The leading equivalent dark density is $\rho_D \simeq \sqrt{[a_0\, \bar{\rho}_b(<r)/(12\pi G r)]} = \kappa\sqrt{[\bar{\rho}_b(<r)\, \rho_{crit}/S]}$.
- Here $\bar{\rho}_b(<r) = 3M_b/(4\pi r^3)$ is the *mean* baryonic density inside the radius r, not the density at the point r.
- $\kappa = 1/3$ exactly if $a_0 = a_{MM}$, and 0.20 for the measured a_0.
- It is an asymptotic rewriting of the assumed RAR, in the stated geometry. It is not an independent law of the density.

A finite reservoir (conditional).

- Suppose we give an isolated system, in addition, a finite dark reservoir, $M_{D,max}$. And suppose we treat its outer field as spherical, with a concentrated baryonic mass M_b.
- Then the adopted RAR exhausts that reservoir at the radius $r_{es} = \sqrt{(GM_b/a_0)}/\ln(1 + 1/q_{max})$, with $q_{max} = M_{D,max}/M_b$.
- Choosing $M_{D,max} = 3\sqrt{3}\cdot M_{b,collected}$ requires a separate rule to say which baryons have been collected.
- **Example from the text.** $M_b = 6\times10^{10}\ M_\odot$ and $1.6\times10^{11}\ M_\odot$ collected give $r_{es} \approx 120$ kpc. At 300 kpc the velocity would be 113 km s^{-1} instead of 177. These numbers assume a full reservoir and only $6\times10^{10}\ M_\odot$ of gravitating baryons within 300 kpc. Other baryons around the galaxy would change them.
- A comparison with the weak lensing of isolated galaxies (Mistele and colleagues 2024) will be possible. But only when the reservoir has been defined independently.

The normalisation.

- The literal value $a_{MM} = cH_0/2 = 3.27\times10^{-10}$ m s^{-2} is 2.73 times the SPARC value. In this record a_0 is taken from the data.
- One can do a *toy* algebraic rescaling, that is, a very simplified one, of the vertical field near the Sun. It gives about 66 $M_\odot\ pc^{-2}$ with the empirical normalisation and 92 with the literal one.
- It is not a self-consistent mass model of the Galaxy, and it is not used as a quantitative test.

7. Problems this reading tries to connect

The aim. We want to understand whether a single interpretation with a hidden sector can connect several dark-matter problems. Today the connections have different statuses: imposed numerical hypotheses, phenomenological fits, qualitative assumptions. Other research programmes also address combinations of these problems. Today this is an *aim* of unification, not four consequences derived from a single model.

Consilience. The case for this reading is that a single interpretation connects problems that are usually treated separately. This case is only as strong as the connections that become quantitative. Today two out of four are.

problem	quantitative status in this record	related approaches
Density ratio between dark matter and baryons	Fixed numerical hypothesis, not derived: 5.196; retrospective consistency check (T_B 0.17-0.23) and registered future test	Asymmetric and mirror dark matter: quantitative abundance relations that depend on the model (Kaplan, Luty and Zurek 2009; Berezhiani 2008)
Cosmological scale of a_0	Phenomenological H(z) fit: lowest χ^2 among three specified one-parameter laws, but inadequate in the c95 reading; the literal normalisation is high by 2.73	Milgrom's link with the cosmological acceleration (Milgrom 1999); Verlinde's de Sitter construction (equivalent scale $cH_0/6$), with different assumptions on the evolution
Absence of conventional direct-detection signals	The absence of non-gravitational interactions with visible matter is assumed; there is no calculation of the detection rates	Hidden and mirror sectors with absent or weak portal interactions
CDM-like cosmology and acceleration relations of galaxies	The CDM behaviour is assumed and the empirical RAR is imposed; their dynamical link is not derived	AeST (Skordis and Złośnik 2021); superfluid dark matter (Berezhiani and Khoury 2015); ΛCDM models of galaxy formation (Ludlow and colleagues 2017); some dissipative hidden-sector models (Chashchina, Foot and Silagadze 2017)

How to read the table. The first column is the problem. The second says how quantitative the connection is in this record. The third lists who else addresses the same problem.

The two quantitative hypotheses examined here are the fixed density ratio and the evolution law H(z). Neither of the two is derived from the hidden-sector construction. The decay analysis of section 4 is one more constraint on a phenomenological model. It does not explain why dark matter is stable.

8. Part 1: what the MM algebra provides for a hidden sector

Where it comes from. This part is taken from an earlier, unpublished draft. The program modules recompute the stated quantities on three test *banks*.

Two words to read the table. In the MM algebra there are *closed modes*: parts of the model closed on themselves, like the cycles of a network. A *declared translation* is a choice, registered in advance, of how to put an idea of the author into the formulas. The *commutator* measures whether two operations can be swapped without changing the result. If it is zero, one does not disturb the other.

statement	status
A declared translation protects closed modes	commutator residuals at numerical precision (largest element below 4×10^{-15}); exact invariance is an algebraic claim under the stated definitions (`ESITO_12`)
Closed modes weigh like all the others	a partition of unity ($\Sigma\varepsilon_i = H$)
Invisibility to our light must be assumed	every closed cycle responds to an edge connection at second order (minimum response 0.0036-0.033)
A whole neutral copy is compatible with weight; protecting only the cycles of one copy fails	`ESITO_13`
The eight-state scaffold of MM 2.29 has two equivalent copies of equal mass	derived in the scaffold (`ESITO_18`)
A conserved common number not carried by photons	forbids decay into photons alone for a pair that carries a non-zero total of that number
Separation of the two lights	obtained on disconnected light networks in the declared class of local operators; checked on 60 random graphs; grouping the channels is not enough (the reading returns, up to 25.5 %); a small link between the photons reopens communication at large distance

How to read some rows.

- *Closed modes weigh like all the others*: in the algebra their weights enter the total sum like those of all the other modes ($\Sigma\varepsilon_i = H$). Reading them as real masses is an interpretation (see below).
- *Invisibility to our light must be assumed*: the algebra alone does not guarantee it. Every closed cycle responds a little, at second order, if it is connected to our network. So invisibility is an assumption.
- *Separation of the two lights*: the two lights stay separate if their networks are disconnected. Grouping the channels is not enough. And a small link between the photons is enough to reopen communication at large distance.

Two warnings.

- **Equal weights in the algebra** do not determine how much matter there is in the universe, and they do not derive $3\sqrt{3}$. Identifying them with real gravitational masses is an interpretation.
- **What was left out.** The quantitative 1 TeV candidate of the earlier draft. It is superseded by the generic reading of section 6.

9. A worked special case: a single common scale factor

The map. A simple case is tried: all the quantities of the hidden sector are obtained from ours with a single factor A.

- Lengths: $\ell_eff = \ell'/A$. Times: $t_eff = t'/A$. Masses: $m_eff = A\, m'$.
- The speed of light c, the Planck constant ħ and the constant α of electromagnetism do not change.
- We define $x = T'_\gamma/T_\gamma$, the ratio between the temperature of the hidden photons and that of ours. Then the effective ratio of the temperatures is Ax.

- From this follow $\rho_{eff} = A^4\rho'$ and $(\sigma/m)_{eff} = A^{-3}(\sigma/m)'$. Here ρ is the density and σ/m says how much the particles collide, per unit of mass.
- To apply the map in the physical world, a rule is also needed for the shared geometry and for the gravitational coupling.

A screening region. An illustrative screening asks for two conditions:

- $(\sigma/m)_{eff} \leq 1\ \mathrm{cm^2\ g^{-1}}$, with an estimate for cold neutral hydrogen. It means that the hidden particles collide little.
- A residual ionised fraction $x_e \leq 0.01$, with an approximate trend of recombination. It means that very little plasma remains.

These choices give $A \gtrsim 850$ and $x \lesssim 520/A^3$. Then the hidden neutrinos and radiation are negligible. This defines a *formal* screening region. It is not a parameter window shown to be viable.

- The agreement with the estimate of Cyr-Racine and Sigurdson concerns the trend. Their formula is itself uncertain by up to a factor of ten.
- Some things are not computed: three-body reactions (important in the whole region; Gurian and colleagues 2022), the hidden nucleosynthesis, ionisation during collisions and collective plasma effects (Lasenby 2020; DeRocco and Giffin 2025).

What comes out.

- We take $A = 1000$ and $x \approx 5.2\times10^{-7}$, the illustrative upper edge. There we find $\eta' \approx 0.023$: about 44 photons per baryon. This value changes within the region.
- A copy with our ratio between baryons and photons does not pass the adopted screening, in the example considered. No range that works with the observations is established.
- The hidden proton reads as $A\cdot m_p$: 0.94 TeV at $A = 1000$. It is an illustrative conversion.

This is why the record keeps the relative constants generic.

10. How this work can turn out wrong

Two quantitative hypotheses have failure thresholds declared in advance, for tests on future data. The dated specifications of the analyses must be completed before looking at the new results.

- **The ratio.** DESI final BAO, Simons Observatory and complementary Planck will be used, analysed as registered, with Σm_ν free. If they give $T_B \geq 9$, the fixed ratio $3\sqrt{3}$ fails in the declared model.
- The sensitivity of this combination of data has not been forecast here.
- We make a Gaussian approximation. A true ratio of 5.30 or 5.10 would nominally reach $T_B = 9$ if the error $\sigma(\omega_c/\omega_b)$ were about 0.035 or 0.032. The real detection power requires a dedicated forecast.
- **The shape of $a_0(z)$.** If the registered global shape test gives $p < 0.0027$, the specified Hubble-rate law fails that test. The test uses a free normalisation for each homogeneous sample, the same for all its bins. And it uses the declared treatment of systematic errors.
- These outcomes concern the specified quantitative hypotheses, within their declared assumptions.
- **Other results would challenge specified versions or assumptions.**
- Evidence of decay would challenge the version with stable cold dark matter.
- The dynamics of cluster mergers could challenge a specified realisation with collisionless halos.
- Microphysics calculations could exclude specified realisations of the hidden sector.
- These are not additional registered failure thresholds.

11. What is not claimed

- **$3\sqrt{3}$ is not derived here.** It is a hypothesis of MM, consistent with the current CMB and BAO data and registered for future data.
- **The RAR is not derived.** Its form and a_0 are taken from the data. The literal normalisation of MM is wrong by a factor of 2.73.
- **The dynamics of the hidden sector is not computed.** That it moves like cold dark matter is assumed. The hidden radiation that does not gravitate is a postulate, without a covariant model.
- **The Bullet cluster is not computed.** A component with almost no collisions is the intended qualitative interpretation. It is not a demonstrated solution of the collision.
- **The limits on decay are nominal and provisional,** and they hold for a single model.
- **No observational confirmation of MM is claimed.**

- **Closeness between numbers is not treated as a derivation, nor as an observational confirmation of MM.**

How the work was done

The working names. In the working records this deposit is called «materia oscura» (folder 29). Some files carry the provisional name «MM 2.46» or the numbers of the working folders. They belong to this record.

Who did what.

- **The author.** The framework and its premises are the author's. His words that started these calculations are in the file `WHERE_THIS_COMES_FROM.md`, with a translation. No result depends on them.
- **Sections 2-6 and 9.** The calculations and the code are by Claude (Anthropic), an AI assistant. The working protocols are in `registrazioni/`, in Italian, kept as they were written. They have SHA-256 hashes of the contents and dated records. A hash identifies the contents of a file; every claim of earlier registration rests on the documented chronology.
- **Cross-checks.** The archive contains the checks of a second AI assistant, Astra (GPT-6, in a Codex session). Their scope, the corrections and the unresolved points are documented in `documenti/note_di_verifica/` (in Italian). Claude re-verified the new numbers with code. These checks are not an independent observational replication, nor a peer review done by people.
- **Section 8 (Part 1).** The derivations are by Astra. Claude checked them with independent code: modules 01 and 06 are Claude's code. The derivations are in `documenti/parte1_algebra/astra/`; the independent checks are in `documenti/parte1_algebra/verifiche/`, both in Italian.
- **Withdrawn statements are recorded,** not hidden. Here are three examples. An early «3σ» for a non-nested comparison. A 3 % limit on condensed hidden matter. A coincidence with 1 + Ω_Λ.
- **No human referee has examined this record.**

The programs and the registrations by section.

- Section 2: `codice/08_cosmology_results.py`, `registrazioni/rapporto_3r3/`. The registration for future data is `registrazioni/REGISTRATION_3SQRT3_EN.md`.
- Section 3: `codice/07_rar_scale_evolution.py`, `registrazioni/a0_redshift/`. The registration for future data is `registrazioni/REGISTRATION_A0_HUBBLE_EN.md`; the working one, in Italian, is `registrazioni/a0_redshift/PREVISIONE_A0_HUBBLE_v2.md`.
- Section 4: `codice/08_cosmology_results.py`, `registrazioni/decadimento/`.
- Section 5: `codice/09_rar_reading.py`.
- Section 8: `codice/01_algebra_sectors.py`, `06_light_separation.py`.
- Section 9: `codice/10_hidden_sector_scaling.py`.

How to redo the calculations.

```
python riproduci.py          # about 1.5 minutes on one CPU
python riproduci.py --full   # also re-runs the slow optimisation of module 01
```

- **What you need:** Python 3.11 or later, with numpy, scipy and sympy. No network access.
- **Where the results go:** `risultati/`.
- **The cosmological minimisations are not redone by this command.**
- Their saved minima are in `dati/risultati_cosmologici/`:
- `corse_6/`: first runs, L0 = ΛCDM, DD = decay with Γ free, DF = Γ = 2.01;
- `profilo/`: profile points, R0, R2;
- `vincolo_3r3/`: D-B;
- `camb_confronto/`: CAMB ΛCDM.
- The scripts are in `codice/corse_cosmologiche/`.
- **To redo them** you need Cobaya 3.6.2, CLASS 3.4.1 (classy) and the Planck 2018 likelihoods (`cobaya-install`). The scripts carry the paths of the original Linux machine (WSL). You must adapt `packages_path` and the path of the DESI data in `desi_dr2_bao.py`. The DESI files are in `dati/desi_dr2/`, identical by SHA-256.
- **The computing time:** one profile point takes 15-25 minutes on two threads.

The contents of the package.

path	what
`README.md`	the archived text

`LEGGIMI_IT.md`	summary in Italian
`WHERE_THIS_COMES_FROM.md`	the author's words, with translation (optional)
`riproduci.py`	reproduction of the included calculations and of the checks on the archived results
`codice/01_*.py`, `06_*.py`	Part 1: algebra of hidden sectors; separation of light
`codice/07_*.py` … `10_*.py`	$a_0(z)$; cosmological results; RAR reading and clusters; one-factor special case
`codice/corse_cosmologiche/`	Cobaya/CLASS scripts used for the minimisations, and the DESI DR2 likelihood
`dati/`	MUSE-DARK III bins (Alcock 2026), mean and covariance of DESI DR2 BAO, saved minima
`risultati/`	JSON and logs of the modules
`registrazioni/`	working protocols (in Italian) and the two registrations in English
`documenti/`	verification notes (in Italian) and the derivations of Part 1
`fonti/SOURCES.md`	references
`IMPRONTE.sha256`	SHA-256 of every file

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **a_0.** The acceleration scale of the RAR: about 1.20×10^{-10} m s^{-2} for nearby galaxies (SPARC).
- **BAO (baryon acoustic oscillations).** A typical distance imprinted in the distribution of galaxies, used as a cosmic ruler.
- **Baryons.** Ordinary matter: mostly protons and neutrons.
- **Bin.** A group of galaxies at similar distances, summarised by a single value.
- **CDM.** Cold dark matter: its particles move slowly.
- **χ^2 (chi-squared).** An error score: the lower it is, the better the model follows the data.
- **Cluster of galaxies.** A large group of galaxies held together by gravity, immersed in very hot gas.
- **CMB (cosmic microwave background).** The oldest light of the universe, measured by the Planck satellite.
- **Consilience.** When a single idea holds together different facts, usually treated separately.
- **Covariant.** Written in a way that is valid for every observer, as relativity requires.
- **Decay.** The transformation of a particle into others; here, of dark matter into dark radiation.
- **ΔN_{eff}.** How much extra radiation there is in the young universe compared with the expected one.
- **Dex.** A factor of ten on a logarithmic scale.
- **Fit.** The calculation that chooses the numbers of a model so that it follows the data as well as possible.
- **Halo.** The cloud of dark matter around a galaxy or a cluster.
- **Hidden sector, "mirror".** Matter with its own particles and its own forces, which shares only gravity with us.
- **ΛCDM.** Standard cosmology: constant dark energy (Λ) and cold dark matter (CDM).
- **Lensing.** The bending of light by the mass it meets.
- **Lifetime.** The typical time before a particle decays; here Γ^{-1}.
- **Likelihood.** The measure of how well a model follows the data. The scores χ^2 and Q go the other way: the lower they are, the better the model follows the data.
- **Phenomenological.** Chosen because it describes the data, not derived from a theory.
- **Prospective and retrospective.** A prospective test uses data still to come; a retrospective one uses data that

already existed.
- **RAR (radial acceleration relation).** The link, point by point, between the acceleration observed in a galaxy and the acceleration of the baryons alone.
- **Redshift (z).** How much the light has been stretched during its journey: the larger z is, the farther away and the further back in time we look.
- **Registration.** A hypothesis written down and dated before seeing the data with which it will be tested.
- **Screening.** Here: the conditions for the hidden sector to collide little and to remain almost without plasma.
- **σ (sigma).** The typical error of a measurement.

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